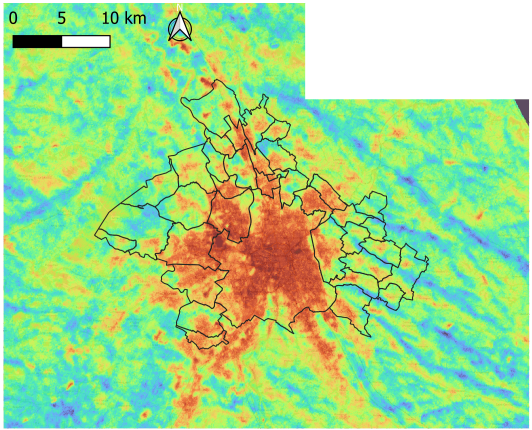
Panoptic segmentation of airborne hyperspectral images for land cover mapping in large urban areas

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November 2, 2026


0
5
10 km

▸ Radiative properties of artificial impermeable surfaces increase **urban heat island effects** [IPCC, 2021].

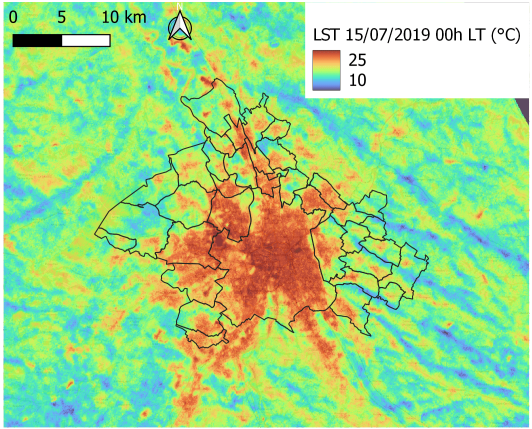


Figure: Land surface temperature measured in Toulouse from the Ecostress mission on 15 July 2019 at midnight. Source: THERMOCITY[1]

[1] Data source: https://www.theia-land.fr/product/thermocity/

Hyperspectral imaging

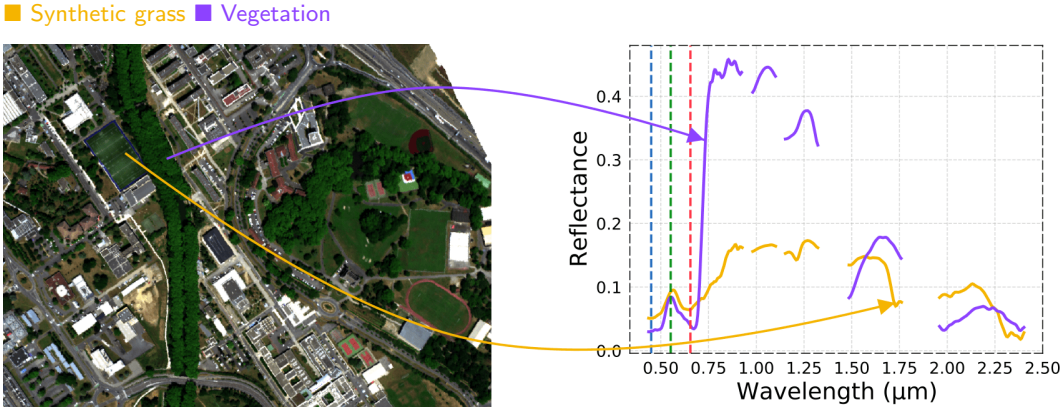


Figure: Hyperspectral airborne image (1m spatial resolution) of ISAE-SUPAERO / ONERA / CREPS.
Source: ONERA

Hyperspectral imaging

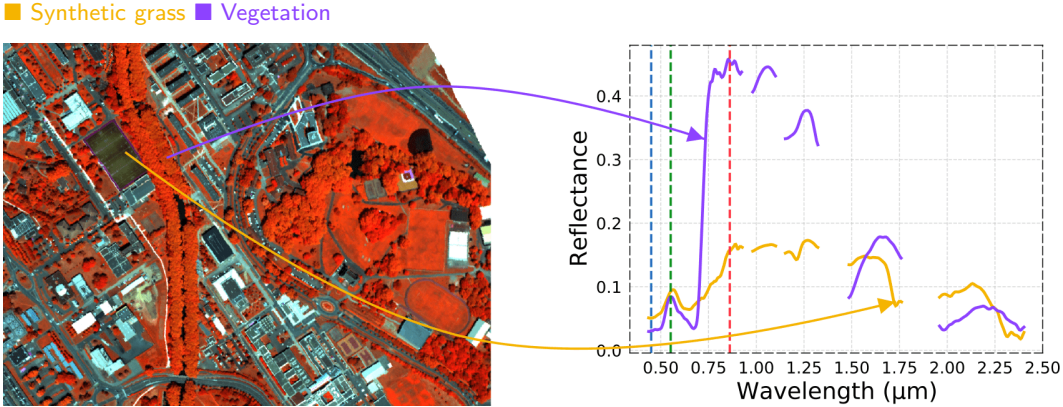


Figure: Hyperspectral airborne image (1m spatial resolution) of ISAE-SUPAERO / ONERA / CREPS.
Source: ONERA

Complexity of the land cover in metropolitan areas



Figure: Parts of a RGB composition of a hyperspectral image over the metropolitan area of Toulouse

Complexity of the land cover in metropolitan areas



Figure: Parts of a RGB composition of a hyperspectral image over the metropolitan area of Toulouse

Complexity of the land cover in metropolitan areas



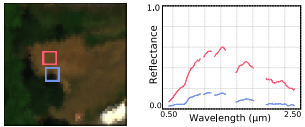
Figure: Parts of a RGB composition of a hyperspectral image over the metropolitan area of Toulouse

Spectral intra-class variability

Physics variability

Caused by **different illumination conditions**

- Pixels in the shadows or in the sunlight,
- Pixels on a flat ground or on a slope.

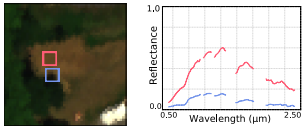


Spectral intra-class variability

Physics variability

Caused by **different illumination conditions**

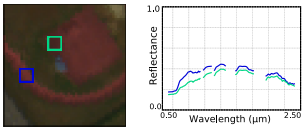
- Pixels in the shadows or in the sunlight,
- Pixels on a flat ground or on a slope.



Intrinsic variability

Caused by **slight variations of the chemical composition / non-lambertian effects**

- Different environmental exposures (*e.g. moss on roofs facing North*),
- Different agings (recent / old asphalt),
- For vegetation, different water / chlorophyll concentrations.

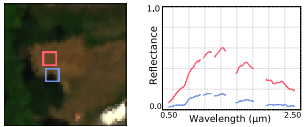


Spectral intra-class variability

Physics variability

Caused by **different illumination conditions**

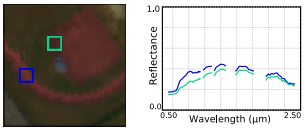
- Pixels in the shadows or in the sunlight,
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Intrinsic variability

Caused by **slight variations of the chemical composition / non-lambertian effects**

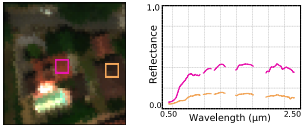
- Different environmental exposures (*e.g. moss on roofs facing North*),
- Different agings (recent / old asphalt),
- For vegetation, different water / chlorophyll concentrations.



Semantic variability

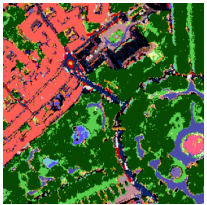
Caused by **large variations of the chemical composition**

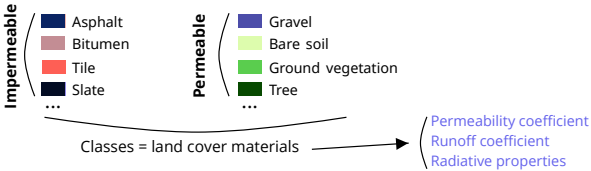
- Different materials gathered in the same semantic class



Land cover mapping via pixel-wise classification of hyperspectral images







Panoptic segmentation

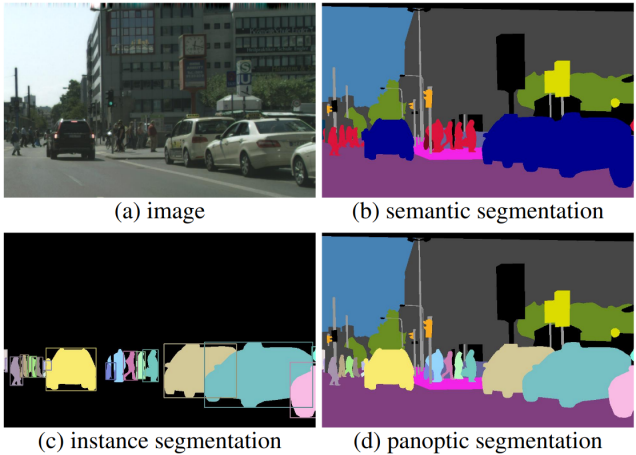
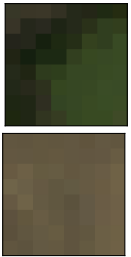


Figure: Figure from Kirillov et al., 2019[2].

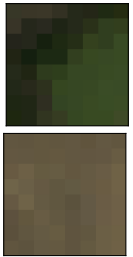
[2]Kirillov, A., He, K., Girshick, R., Rother, C., & Dollár, P. (2019, June). Panoptic segmentation. In 2019 IEEE/CVF conference on computer vision and pattern recognition (CVPR) (pp. 9396-9405). Ieee.

Key issues in the processing of hyperspectral images



(a) 8×8 pixels ($8m \times 8m$)

Key issues in the processing of hyperspectral images

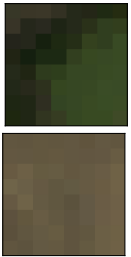


(a) 8×8 pixels
$(8m \times 8m)$



(b) 16×16 pixels
$(16m \times 16m)$

Key issues in the processing of hyperspectral images



(a) 8×8 pixels
$(8m \times 8m)$

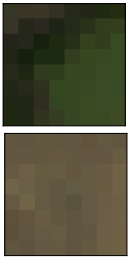


(b) 16×16 pixels
$(16m \times 16m)$



(c) 32×32 pixels
$(32m \times 32m)$

Key issues in the processing of hyperspectral images



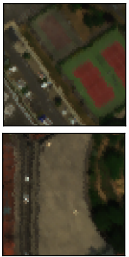
(a) 8×8 pixels ($8m \times 8m$)



(b) 16×16 pixels ($16m \times 16m$)



(c) 32×32 pixels ($32m \times 32m$)



(d) 64×64 pixels ($64m \times 64m$)

Combination of hyperspectral images with vector land-use data

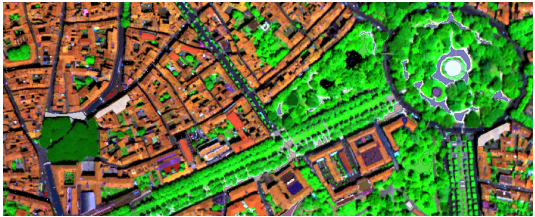


Figure: Ground truth annotations

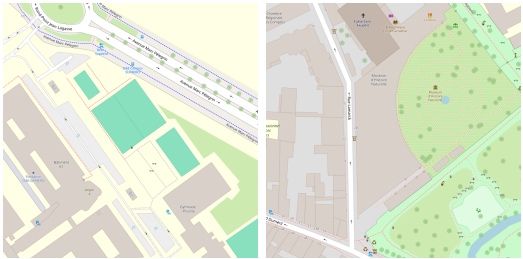


Figure: Open Street Map vectorial data

Research question

How to combine airborne hyperspectral images with land use vectorial data to produce ready-to-use land cover maps through panoptic segmentation ?

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Key issues to be resolved

- High spatial and spectral dimensions of hyperspectral images.
- Gap between available ground truth and targeted output.

Questions?

References

- Hyperspectral land cover data set: `https://www.toulouse-hyperspectral-data-set.com/`
- Kirillov, A., He, K., Girshick, R., Rother, C., & Dollár, P. (2019, June). **Panoptic segmentation**. In 2019 IEEE/CVF conference on computer vision and pattern recognition (CVPR) (pp. 9396-9405).
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- Audebert, N., Le Saux, B., & Lefèvre, S. (2017, July). **Joint learning from earth observation and openstreetmap data** to get faster better semantic maps. In 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (pp. 1552-1560). IEEE.

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